

Limits of rational polynomials

Key Points

To find limit of

$$\lim_{x \rightarrow a} \frac{N(x)}{D(x)}$$

Problems with limits arise from

$$“\frac{0}{0}, \frac{\infty}{\infty}, \frac{\infty}{0}”$$

As $x \rightarrow 0$ we divide by the smallest power of x . As $x \rightarrow \infty$ we divide by the largest power of x

Basic Skills

Q1. For the function $y = \frac{1}{x}$

- a) Sketch the graph
- b) State the two asymptotes of the graph/function, distinguishing between them
- c) State $\lim_{x \rightarrow \infty} \frac{1}{x}$
- d) Why is it not possible to state $\lim_{x \rightarrow 0} \frac{1}{x}$

Q2. For the function $y = \frac{4x^2 - x - 2}{x^2}$

- a) Show that $y = 4 - \frac{1}{x} - \frac{2}{x^2}$
- b) Find the value of y as $x \rightarrow \infty$
- c) Find the value of y as $x \rightarrow 0$

Competency Skills

Q3. State the type of problem with evaluating the following limits

a) $\lim_{x \rightarrow 0} \frac{x^2 - x}{x}$ b) $\lim_{x \rightarrow \infty} \frac{x^3 + x^2}{4x^3}$ c) $\lim_{x \rightarrow \infty} \frac{x^3 + x^2}{4e^{-x}}$

Q4. Find the following limits

a) $\lim_{x \rightarrow 0} \frac{x^2 - x}{x}$ b) $\lim_{x \rightarrow \infty} \frac{x^3 + x^2}{4x^3}$ c) $\lim_{x \rightarrow \infty} \frac{x^3 - 2x^4 + x + 2}{x^4}$ d) $\lim_{x \rightarrow 0} \frac{x+2}{x^2 - 1}$

e) $\lim_{x \rightarrow \infty} \frac{(x+2)^2}{2x(x+1)}$ f) $\lim_{x \rightarrow \infty} \frac{(3x-1)^2}{2x^3}$ g) $\lim_{y \rightarrow \infty} \frac{5y^3 - 7y + 5}{2y^2 + 5}$

Q5. Find $\lim_{x \rightarrow 0} \frac{(2x - \frac{2}{x})^2}{x^2}$

Advanced Skills

Q6. Find $\lim_{x \rightarrow 9} \frac{(x-9)^2}{81-x^2}$

Q7. Find $\lim_{x \rightarrow 0} \frac{\sin x}{\cos x}$, stating why you can work out this limit even though it is not a polynomial

Q8. A function is given by $f(x) = \begin{cases} \frac{2x-5}{6x^2+5} & \text{if } x \leq 0 \\ \frac{6x^2+5}{2x-5} & \text{if } x > 0 \end{cases}$

- a) Evaluate the limit $\lim_{x \rightarrow \infty} f(x)$
- b) Evaluate the limit $\lim_{x \rightarrow -\infty} f(x)$
- c) Evaluate the limit $\lim_{x \rightarrow 0} f(x)$
- d) If we change $f(x) = 2$ if $x > 0$ does the answer to c) change

Extension

Q9. Technically a non-infinite limit, such as $x \rightarrow 0$, must have the same value approaching from both directions. State a function where approaching from the below and approaching from above are not equal